EXTENDING OF MENDELIAN GENETICS: Lec.2

-Many geneticists have extended Mendelian principles to pattern of inheritance more complex than that describes by Mendel.

*ALLELE INTERACTIONS (INCOMPLETE DOMINANCE)

A type of inheritance in which the offspring shows characteristics intermediate between two extreme parental features (neither allele is fully dominant). This phenomenon is known as incomplete dominance.

-For example in across between a homozygous red flowering snapdragon (RR) and a homozygous white flowering snapdragon (R-R-) produces heterozygous offspring with pink flowers. This is not an example that supports the blending theory of inheritance because when F1 allowed to self pollinate the F2 generation has a phenotypic ratio of 1 red, 2pink, 1white flowered plant (parental phenotypes reappeared in the F2).

P1 RR R- R-

G1 R R-

F1 RR- 100% pink

RR- RR-

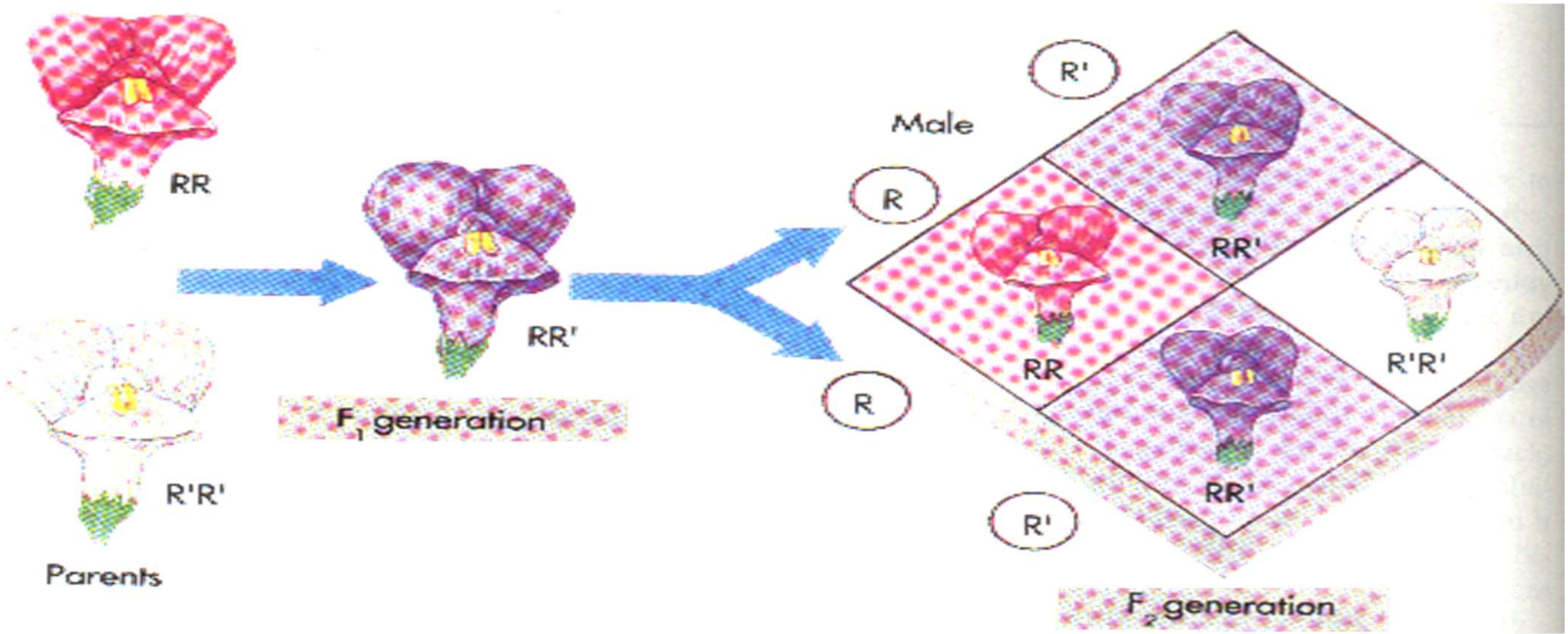
R R-

R RR RR-

F2 R- RR- R- R-

1 : 2 : 1

red : pink : white


RR
RR'
R'R'
Parents
F1 generation
Male
R'
R
R
R'
RR'
RR
R'R'
RR'
F2 generation

*Co-dominant:

Alleles may act in a co-dominant manner, in which both alleles are apparently fully expressed in the phenotype of heterozygote. An individual with the blood type AB is exhibit co-dominant characteristics in which two different types of glycoprotein appear on the RBCs.

$$I^A I^B \qquad I^A I^B$$

$$I^A I^A,\ I^A I^B,\ I^A I^B,\ I^B I^B$$

1: 2: 1

*MULTIPLE ALLELES (SEVERAL ALLELES FOR ONE TRAIT):

Some times there are more than two alleles for a given chromosomal locus, in which case a trait is controlled by multiple alleles, in rabbits there are four alleles for coat color.

$$C > c^{ch} > c^{h} > c^{a}$$

Agouti, chinchilla, hemalaya, albino.

The ABO blood groups in humans are controlled by three alleles and there are four possible phenotypes: A, B, AB, and O.

-A person's red blood cells may be coated with one glycoprotein surface antigen (type A or B), with both (type AB) or with neither (type O). Matching compatible blood groups is critical for blood transfusions. A person with type A, has anti-B antibodies, and one with type B has anti-A antibody, and blood cells clump, or agglutinate when they are mixed with antibodies against the antigens they bear.

-This agglutination can cause destruction of blood cells, tissue breakdown, fever and death. People with type AB blood, having neither anti-A nor anti-B antibodies, can accept blood of any type while people with type O blood, having no A or B antigens on their blood cells, can donate to any one.

Can someone with blood type AB have a child with Type O?

Yes .If the child's RBCs lacking H antigen (When inherited two recessive alleles hh) due to mutations in FUT1 gene, both A and B antigens failed to express because the H substrate is a precursor for A and B antigens that's why such a child will at least shows the type O blood group functionally, but with different genotype. This rare condition is known as Bombay phenotype.

PLEIOTROPY (A GENE THAT CONTROLS MANY TRAITS):

The term pleiotropy is used to describe a gene that affects more than one characteristic of the individual. Individuals with Marfan syndrome tend to be tall, and thin with long legs,arms,and fingers. They are nearsighted and the wall of aorta is weak, causing it to enlarge and eventually to split. All of these effects are caused by an inability of agene to produce normal connective tissue.

POLYGENIC INHERITANCE AND ENVIRONMENT:

Some traits, such as size or height, weight, shape, color, metabolic rate and behavior are not the result of interaction between one, two, or even several genes; instead they are the cumulative result of the combined effects of many genes and environment. The phenomenon is known as polygenic inheritance.

-The height is controlled by multiple genes. It's believed that up to 70% of a person's height is determined by genetic and the environmental factors including nutritional status, healthcare and diseases determining the rest.

-Human skin color and intelligence are believed to be under the influence of a similar kind of genetic or environmental factors.

Pattern of inheritance:

- Many human traits are inherited. Hair and eye color, facial characteristic, aspect of personality all are passed from parents to children. Some major clinical disorders are also inherited, and it is important for investigators to be able to study them, both to assess the likelihood that parents at risk will produce affected children and to seek future cures

-To study how a human trait is inherited, investigators look at the results of crosses that have already been made; they study family histories, called pedigrees.

- By studying which relatives exhibit a trait, it is often possible to say if the gene producing the trait is:

sex linked or autosomal and to determine if the trait's phenotype is dominant or recessive and which individuals are homozygous and which are heterozygous for the allele specifying the trait.

Pedigree Symbols